

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2015

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number** (not your name) on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. Suppose that

$$1 \rightarrow A \rightarrow B \xrightarrow{g} C \rightarrow 1$$

is an exact sequence of groups, where A has order 85 and C has order 9.

- (a) Let S be a 3-Sylow subgroup of B . Show that g maps S isomorphically to C .
- (b) Show that $|\text{Aut}(A)| = 64$ (where Aut denotes the automorphism group of a group).
- (c) Show that B is abelian.

2. Let $n \geq 2$ and consider the $n \times n$ complex matrices defined by

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \cdots & & & & \cdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix} \quad N = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \cdots & & & & \cdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

(these are Jordan blocks with an extra 1 added in the upper right corner or the lower left corner).

- (a) Find the characteristic polynomial and the minimal polynomial of M . (*Hint*: The vectors $(1, \zeta, \zeta^2, \dots)^T$, where $\zeta^n = 1$, could be useful.)
- (b) Find the characteristic polynomial and the minimal polynomial of N . (*Hint*: N^2 is the same as the square of the Jordan block obtained by removing the 1 in the upper right corner.)

3. Let $R = \mathbb{C}[X, Y]$ and let $M = (X, Y)$ be the ideal of R generated by X and Y .

- (a) Show that the map

$$M \otimes_R M \rightarrow \mathbb{C}$$

defined by

$$\sum_i f_i \otimes g_i \mapsto \sum_i \frac{\partial f_i}{\partial X}(0, 0) \frac{\partial g_i}{\partial Y}(0, 0)$$

is a well-defined map of vector spaces over $\mathbb{C}$.

- (b) Show that $Z = X \otimes Y - Y \otimes X \neq 0$ in $M \otimes_R M$.
- (c) Show that there is an element $r \neq 0$ in R such that $rZ = 0$ in $M \otimes_R M$.

4. Let R be a commutative ring with $1 \neq 0$.

- (a) Let $I_1, I_2, \dots, I_n$ be ideals of R and let P be a prime ideal of R such that

$$I_1 \cap I_2 \cap \cdots \cap I_n \subseteq P.$$

Show that $I_i \subseteq P$ for some i .

(b) A *minimal prime ideal* of R is a prime ideal m of R such that whenever Q is a prime ideal with $Q \subseteq m$ then $Q = m$. Show that R contains at least one minimal prime ideal (you may use Zorn's Lemma, or something equivalent).

(c) Suppose there are minimal prime ideals $m_1, m_2, \dots, m_n$ of R such that

$$m_1 \cap m_2 \cap \dots \cap m_n = (0).$$

Show that $\{m_1, m_2, \dots, m_n\}$ is the set of all minimal prime ideals of R .

5. Let $\mathbb{Q} = K_0 \subset K_1 \subset K_2 \subset \dots \subset K_n$ be a sequence of fields such that K_{i+1}/K_i is Galois of degree 3 for each $i \geq 0$. Show that $\mathbb{Q}(\sqrt[3]{2})$ is not contained in K_n .
6. Let $\rho : G \rightarrow \mathrm{GL}_n(\mathbb{C})$ be a representation of the finite group G . Define a representation $\tilde{\rho} : G \times (\mathbb{Z}/2\mathbb{Z}) \rightarrow \mathrm{GL}_{2n}(\mathbb{C})$ by

$$\tilde{\rho}(g, 0) = \begin{pmatrix} \rho(g) & 0 \\ 0 & \rho(g) \end{pmatrix}, \quad \tilde{\rho}(g, 1) = \begin{pmatrix} 0 & \rho(g) \\ \rho(g) & 0 \end{pmatrix}.$$

- (a) Show that $\tilde{\rho}$ is a representation of $G \times (\mathbb{Z}/2\mathbb{Z})$.
- (b) If π is an irreducible representation of G , then $\pi'(g, a) = \pi(g)$ defines an irreducible representation π' of $G \times (\mathbb{Z}/2\mathbb{Z})$. Show that the number of times that π occurs in ρ equals the number of times that π' occurs in $\tilde{\rho}$.